

35) Let $A = \begin{bmatrix} -2 & -7 & -9 \\ 2 & 5 & 6 \\ 1 & 3 & 4 \end{bmatrix}$. Find the third column of A^{-1} without computing the other columns.

Solution:

$$\begin{aligned}
 & \begin{bmatrix} -2 & -7 & -9 & | & 0 \\ 2 & 5 & 6 & | & 0 \\ 1 & 3 & 4 & | & 1 \end{bmatrix} \xrightarrow{\text{Row 2} = \text{Row 2} + \text{Row 1}} \begin{bmatrix} -2 & -7 & -9 & | & 0 \\ 0 & -2 & -3 & | & 0 \\ 1 & 3 & 4 & | & 1 \end{bmatrix} \\
 & \xrightarrow{\text{Row 3} = \text{Row 3} - \frac{1}{2} \times \text{Row 1}} \begin{bmatrix} -2 & -7 & -9 & | & 0 \\ 0 & -2 & -3 & | & 0 \\ 0 & -1/2 & -1/2 & | & 1 \end{bmatrix} \xrightarrow{\text{Row 1} = \text{Row 1} \times (-1/2)} \begin{bmatrix} 1 & 7/2 & 9/2 & | & 0 \\ 0 & -2 & -3 & | & 0 \\ 0 & -1/2 & -1/2 & | & 1 \end{bmatrix} \\
 & \xrightarrow{\text{Row 2} = \text{Row 2} \times (-1/2)} \begin{bmatrix} 1 & 7/2 & 9/2 & | & 0 \\ 0 & 1 & 3/2 & | & 0 \\ 0 & -1/2 & -1/2 & | & 1 \end{bmatrix} \xrightarrow{\text{Row 3} = \text{Row 3} \times (-2)} \begin{bmatrix} 1 & 7/2 & 9/2 & | & 0 \\ 0 & 1 & 3/2 & | & 0 \\ 0 & 1 & 1 & | & -2 \end{bmatrix} \\
 & \xrightarrow{\text{Row 3} = \text{Row 3} - \text{Row 2}} \begin{bmatrix} 1 & 7/2 & 9/2 & | & 0 \\ 0 & 1 & 3/2 & | & 0 \\ 0 & 0 & -1/2 & | & -2 \end{bmatrix} \xrightarrow{\text{Row 3} = \text{Row 3} \times (-2)} \begin{bmatrix} 1 & 7/2 & 9/2 & | & 0 \\ 0 & 1 & 3/2 & | & 0 \\ 0 & 0 & 1 & | & 4 \end{bmatrix} \\
 & \xrightarrow{\text{Row 1} = \text{Row 1} - 9/2 \times \text{Row 3}} \begin{bmatrix} 1 & 7/2 & 0 & | & -18 \\ 0 & 1 & 3/2 & | & 0 \\ 0 & 0 & 1 & | & 4 \end{bmatrix} \xrightarrow{\text{Row 2} = \text{Row 2} - 3/2 \times \text{Row 3}} \begin{bmatrix} 1 & 7/2 & 0 & | & -18 \\ 0 & 1 & 0 & | & -6 \\ 0 & 0 & 1 & | & 4 \end{bmatrix} \\
 & \xrightarrow{\text{Row 1} = \text{Row 1} - 7/2 \times \text{Row 2}} \begin{bmatrix} 1 & 0 & 0 & | & 3 \\ 0 & 1 & 0 & | & -6 \\ 0 & 0 & 1 & | & 4 \end{bmatrix}
 \end{aligned}$$

$\therefore$ The third column of A^{-1} is $\begin{bmatrix} 3 \\ -6 \\ 4 \end{bmatrix}$ $\square$

37) Let $A = \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 5 \end{bmatrix}$. Construct a 2×3 matrix C (by trial and error) using only 1, -1 and 0 as entries, such that $CA = I_2$. Compute AC and note that $AC \neq I_3$.

Solution:

$$\begin{aligned}
 & \text{Let } C = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{bmatrix} \quad CA = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 3 \\ 1 & 5 \end{bmatrix} \\
 & \quad \quad \quad \text{(By defn. of matrix multiplication)} \\
 & C \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = 1 \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} + 1 \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} + 1 \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \quad \text{(By defn. of } A_2) \\
 & = \begin{bmatrix} a_1 + b_1 + c_1 \\ a_2 + b_2 + c_2 \end{bmatrix} \quad \text{(By Vector addition and Scalar multiplication)}
 \end{aligned}$$